

Security Incident Report

Windows Authentication Event Investigation

Analyst Information

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Role: Future SOC Analyst | Cloud Security Professional

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Case ID: SOC-WIN-2026-001

Severity: Medium

Status: Closed – Investigation Completed

Section 1: Identify the Security Events Involved

The primary security events involved in this investigation were Windows Security Event IDs 4625 (Failed Logon) and 4624 (Successful Logon).

Event ID 4625 indicates that an authentication attempt failed because valid credentials were not provided. Event ID 4624 indicates that a user successfully authenticated to the Windows system.

During the investigation, multiple Event ID 4625 entries were observed within a short time frame, followed by a successful Event ID 4624 authentication event. This sequence is commonly associated with password entry errors, password guessing activity, or brute-force authentication attempts.

The Windows Security Log was analyzed using Event Viewer. Authentication events were filtered and reviewed to identify suspicious patterns and reconstruct the sequence of events.

Security Events Identified

Event ID	Description	Security Relevance
4624	Successful Logon	User Authentication
4625	Failed Logon	Authentication Failure

MITRE ATT&CK Mapping

Technique	ATT&CK ID
Brute Force	T1110
Password Guessing	T1110.001
Valid Accounts	T1078

The observed authentication activity aligns with techniques commonly used by adversaries attempting to gain unauthorized access to systems.

Section 2: Incident Investigation and Analysis

The investigation began after reviewing Windows Security Logs to identify authentication-related activity. Event Viewer was used to access and analyze Windows Security Event Logs.

The cybersecurity analyst filtered Security Logs for Event ID 4625 and identified multiple failed authentication attempts occurring within a short period. Further analysis revealed that the authentication failures generated Windows Security Audit Failure events.

The analyst reviewed the event details and identified authentication failure status code 0xC000006D, which indicates invalid credentials were used during the authentication attempt. The investigation focused on determining whether these failures were isolated user mistakes or indicative of malicious activity.

To establish a timeline of events, the analyst filtered and reviewed Event ID 4624 entries. Event correlation revealed that several failed authentication attempts occurred before a successful authentication event.

Event Timeline

Time	Event ID	Description
18:12:41	4625	Failed Logon
18:12:44	4625	Failed Logon
18:12:45	4625	Failed Logon
18:13:26	4624	Successful Logon

Correlation Analysis

The sequence of events indicates:

Failed Logon (4625)

↓

Failed Logon (4625)

↓

Failed Logon (4625)

↓

Successful Logon (4624)

This pattern is frequently investigated within Security Operations Centers because it may indicate password guessing activity, brute-force attempts, or user authentication errors.

The analyst reviewed available evidence and found no indicators of privilege escalation, malware execution, persistence mechanisms, lateral movement, or account compromise. No unauthorized changes to user accounts, security policies, or system configurations were identified.

Based on the evidence collected, the investigation concluded that the activity was generated within a controlled lab environment and represented authentication testing rather than a confirmed security breach. However, the observed event sequence accurately reflects authentication patterns commonly associated with credential attacks.

Section 3: Risk Assessment

Risk Rating

Medium

Justification

Repeated authentication failures can indicate attempts to gain unauthorized access through password guessing or brute-force techniques. Although the activity occurred in a controlled lab environment, similar patterns in a production environment would warrant further investigation and monitoring.

Potential Impact

- Unauthorized Account Access
- Credential Compromise
- Privilege Abuse
- Security Policy Violations
- Increased Attack Surface

Business Impact

If similar activity occurred within a production environment and was not detected, an attacker could potentially gain access to valid user accounts and use those credentials to access corporate resources.

Section 4: MITRE ATT&CK Analysis

T1110 – Brute Force

Adversaries may attempt to gain access by systematically trying passwords against user accounts.

Evidence: Multiple Event ID 4625 authentication failures were observed.

T1110.001 – Password Guessing

Adversaries may attempt to guess passwords for valid accounts.

Evidence: Repeated authentication failures occurred before successful authentication.

T1078 – Valid Accounts

Adversaries may use valid credentials to access systems and services.

Evidence: A successful Event ID 4624 authentication event occurred after failed authentication attempts.

Section 5: Recommendations

The following security controls are recommended to reduce the likelihood of credential-based attacks:

Multi-Factor Authentication (MFA)

Require users to provide a second authentication factor in addition to a password.

Account Lockout Policy

Automatically lock accounts after a defined number of failed authentication attempts.

Password Security

Implement strong password complexity requirements and prohibit password reuse.

Security Monitoring

Continuously monitor Event IDs 4624 and 4625 and generate alerts for excessive authentication failures.

SIEM Integration

Forward Windows Security Logs to a centralized SIEM platform for real-time monitoring and incident detection.

User Awareness

Educate users on password security best practices and credential protection.

Final Assessment

The Windows Event Log investigation successfully identified and analyzed authentication-related security events within Windows Security Logs. Event correlation techniques were used to reconstruct the sequence of authentication activity and assess potential security implications.

The investigation demonstrated practical SOC analyst skills including Windows Event Log analysis, authentication monitoring, event correlation, threat detection, MITRE ATT&CK mapping, risk assessment, and incident reporting.

No evidence of compromise was identified; however, the event pattern accurately reflects credential attack behavior commonly investigated by Security Operations Center analysts.